

Chapter 1

**SMARTALYZE: AN AI-ASSISTED DATA VISUALIZATION AND ANALYSIS
PLATFORM**



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CHAPTER 1

INTRODUCTION

Background of Study

Globally, the explosion of digital information has made data literacy and data management critical competencies across industries and academic disciplines. Organizations and individuals worldwide increasingly rely on data-driven insights to support decision-making, yet the gap between raw data and meaningful analysis remains a persistent challenge, particularly for non-technical users who lack programming expertise [1]. In fact, data preparation alone is estimated to consume up to 80% of a data analyst's time, underscoring the urgent need for automated, accessible preprocessing solutions [2]. The global shift toward 'Data Democracy' emphasizes that the power of analytics should not be limited to IT experts but should be accessible to all stakeholders in a community [3].

In modern data science, data cleaning and preprocessing are foundational steps that directly influence the quality and reliability of analytical outcomes [4]. Traditional preprocessing workflows are often repetitive, time-consuming, and error-prone, especially for users without technical expertise [5]. This challenge has led to a growing interest in automated data cleaning and analysis tools that can bridge the gap between raw data and actionable insights without requiring advanced coding skills [6].

Automated systems have been proposed to democratize access to data science by simplifying common tasks such as handling missing values, resolving inconsistencies, and applying machine learning models to structured data [7]. Recent research in Human-in-the-loop (HITL) analytics emphasizes that for automated tools to be effective for non-experts, they must provide transparent "change profiles" that explain why certain data points were altered [8].

Furthermore, state-of-the-art frameworks like AutoMind (2026) demonstrate that multi-agent AI architectures can now translate high-level goals into end-to-end analytical workflows, effectively removing the technical barriers that previously restricted data-driven decision-making to expert programmers [9]. These studies highlight both the feasibility and the importance of building systems that make data preparation more accessible to non-technical users.

An example of this is seen in the United Kingdom, where the government has prioritized 'Data-Driven Decision Making' through its National Data Strategy. By automating data cleaning and processing within public health sectors, they significantly reduced the technical burden on non-IT staff, allowing healthcare providers to focus on analysis rather than manual data preparation [10]. Similarly, in Singapore, the 'Smart Nation' initiative utilizes automated AI frameworks to ensure that data insights are accessible to citizens and local businesses, fostering a culture of data literacy across various demographic groups [11].

In the Philippines, the national push toward a digital economy—outlined in the Philippine Development Plan (PDP) 2023–2028 and aligned with the Department of Science and Technology's (DOST) Harmonized National R&D Agenda—has intensified demand for practical tools to bridge the digital skills gap [12]. Furthermore, the integration of data-driven governance is a primary pillar of the National Cybersecurity Plan and the broader e-Governance Act, which mandates the digitization of local government and academic services [13].

Based on preliminary observations and local assessments within the province, students, researchers, and small business owners at institutions such as Davao del Norte State College regularly work with structured data for academic outputs and business operations, yet they face critical operational bottlenecks. It has been observed that academic researchers and students routinely lose valuable time to tedious, manual data-cleaning processes, frequently struggling with missing values and formatting inconsistencies that compromise the integrity of their analysis. Furthermore, local assessments

indicate that small business owners encounter steep learning curves and prohibitive subscription costs when attempting to deploy industry-standard analytics tools. Due to this limited access to both specialized technical expertise and affordable commercial software, these stakeholders are often left with unoptimized data, ultimately leading to delayed academic completions and subjective, unscientific business decisions.

This study aligns with the United Nations Sustainable Development Goals (SDGs) 4 (Quality Education) and 9 (Industry, Innovation, and Infrastructure), as Smartalyze aims to enhance data literacy, promote inclusive access to analytical tools, and foster technological innovation [14]. Furthermore, this project supports the DNSC Research, Development, and Extension (RDE) agenda by contributing a locally relevant, practical system that empowers the DNSC community and the broader Davao del Norte region. The proponents are enrolled at the Institute of Computing of Davao del Norte State College in Panabo City, Davao del Norte, and this study directly addresses the data-preparation challenges faced by stakeholders in the locality who currently lack access to affordable, user-friendly data-analysis tools [15].

To address these challenges, Smartalyze is proposed as an AI-assisted, web-based data-cleaning and analysis system that integrates rule-based preprocessing with basic machine learning to detect trends and handle anomalies. By providing guided suggestions and intuitive visualizations, Smartalyze aims to empower users such as students, researchers, and small business owners to derive meaningful insights from their data without requiring advanced technical skills.

Objectives of the Study

The main objective of this study is to develop Smartalyze, an AI-assisted, web-based data-cleaning and analysis system that simplifies data preparation and insight generation for non-technical users.

Specifically, the study aims to accomplish the following objectives:

1. Develop a web-based data ingestion module that allows users to upload multimodal data sources, including structured datasets in CSV, Excel, and JSON formats, as well as image-based documents (e.g., PNG, JPG) for automated data extraction.
2. Implement an automated data cleaning engine that utilizes rule-based techniques to detect and remediate common data quality issues, ensuring the integrity of the dataset through:
 - 2.1. Imputation or removal of missing values,
 - 2.2. Identification and elimination of duplicate records, and
 - 2.3. Standardization of inconsistent data types.
 - 2.4 Implement Optical Character Recognition (OCR) capabilities to extract text from images for dataset conversion.
3. Implement machine learning-based features to provide intelligent suggestions for data structuring and basic trend analysis, including grouping and summarization.
4. Design a visualization dashboard that generates summarized insights and enables users to interpret trends and patterns through interactive charts.
5. Enable a data export functionality that allows users to download cleaned and processed datasets in CSV, Excel, and JSON formats for academic and business decision-making.

Significance of the Study

The study introduces Smartalyze, an AI-assisted, web-based data-cleaning and analysis system designed to simplify the preparation, analysis, and interpretation of datasets for non-technical users. By automating routine data preprocessing tasks, detecting anomalies, and providing actionable insights, this

system empowers users to make informed, data-driven decisions. The following are the study's main beneficiaries:

Students. Smartalyze enables students to efficiently clean and analyze their datasets for academic projects or research, reducing manual work and improving the accuracy and reliability of their analyses. It also helps them understand patterns and trends in data without needing advanced technical skills.

Small Business Owners. The system allows business owners to upload sales, inventory, or financial data and receive guided insights on trends, anomalies, and performance metrics. This reduces reliance on professional data analysts and supports timely, informed business decisions.

Researchers and Educators. Smartalyze provides a tool that facilitates teaching and learning in data-related fields. Researchers can use the system to quickly process experimental data, while educators can demonstrate data cleaning and analysis concepts in a practical, hands-on manner.

Future Developers. The project lays a foundation for further development of intelligent, user-friendly data analysis tools. Future researchers or developers can build upon Smartalyze to integrate advanced analytics, predictive models, or additional automation features for broader applications.

Scope and Limitation

This study develops and evaluates Smartalyze, an AI-assisted, web-based data-cleaning and analysis system designed to simplify data preparation and analysis for non-technical users. The system allows users to upload datasets in common formats such as CSV, Excel, or JSON. Once uploaded, the system validates the data for basic integrity and automatically detects issues such as missing values, duplicates, and inconsistent data types. It then provides guided suggestions for cleaning and structuring the data, including operations such as grouping, sorting, and summarization, preparing the dataset for further analysis.

After data preprocessing, Smartalyze applies basic machine learning techniques to detect trends and anomalies within the dataset. The system generates summarized insights and visualizations, such as tables and charts, that help users easily interpret patterns and identify unusual data points. Users can also export the cleaned and processed datasets for further use, enabling data-driven decision-making in academic, research, or business contexts. The workflow emphasizes usability and accessibility, targeting students, researchers, and small business owners who may lack advanced technical data analysis skills.

The study is limited to structured datasets with clearly defined rows and columns and does not support unstructured data such as free-text documents. Image-based tabular data is only accepted through the OCR module. The anomaly detection and trend analysis features rely on simple rule-based logic and basic machine learning models, so predictions are probabilistic and may not always capture complex patterns. The system analyzes only datasets that are uploaded by the user and does not perform real-time monitoring or live data tracking. Additionally, very large datasets may experience slower processing times, and the current prototype does not include advanced AI explanations or integration with external analytics platforms.

Review of Related Literature and Works

Related Literature

The increasing volume and complexity of data in modern enterprises have spurred research into automated data preprocessing and analysis tools [16]. A growing body of literature addresses the challenge of making data science accessible to non-technical users by eliminating the need for manual coding or specialized expertise [17]. This section reviews the key themes, gaps, and relevant works that inform the design of Smartalyze.

1. Data Analytics and Data-Driven Decision Making

The contemporary academic and institutional landscape increasingly relies on data-driven decision-making to optimize performance and identify operational insights. This paradigm shift marks a clear transition away from a reliance on subjective intuition or anecdotal evidence toward organizational choices that are strictly grounded in objective dataset metrics. However, extracting real value from these datasets often requires advanced analytical competencies that non-technical users—such as local students, educators, and business researchers—frequently do not possess.

Literature emphasizes that when analytical frameworks require specialized scripting or complex database environments, non-technical users experience structural barriers, causing notable frustration and frequently leading them to abandon data-driven approaches entirely [18]. To bridge this gap, modern research heavily advocates for "data democratization," which is defined as the intentional broadening of data access and analytical tools to non-specialist stakeholders within an organization [19]. This movement emphasizes the deployment of self-service analytics interfaces that minimize cognitive load, effectively allowing individuals without programming backgrounds to comfortably cleanse, visualize, and interpret data trends to make informed decisions [20].

2. Data Cleaning and Data Preprocessing

Research on automated data preprocessing consistently highlights the bottleneck posed by manual data cleaning. Statistical evidence indicates that data preparation accounts for nearly 80% of the analytical lifecycle, underscoring that operations such as handling missing values and normalizing inconsistent formats are foundational yet time-intensive and prone to human error [21].

Furthermore, contemporary evaluations demonstrate that Python-based automation pipelines can significantly accelerate these workflows, reducing manual effort and ensuring reproducible results [22]. Recent benchmark studies

on large real-world datasets confirm that these automated tools can scale effectively without sacrificing data quality [23]. To optimize these pipelines, recent studies emphasize that integrating automated anomaly detection algorithms helps identify structural data irregularities early, dramatically reducing the manual scripting required by end-users [24]. These works collectively establish the feasibility and value of automating data cleaning, forming the theoretical basis for Smartalyze’s rule-based engine.

3. Missing Data Handling Techniques

The integration of machine learning (ML) into data cleaning has received growing scholarly attention. New frameworks demonstrate that ML methods can effectively identify and resolve quality issues, particularly in anomaly detection and missing-value imputation [25]. Recent tools, such as *cleanTS*, extend this approach to time-series data, demonstrating that systems that leverage statistical models can perform domain-specific cleaning with minimal user input [26]. Smartalyze draws from these studies by incorporating basic ML models for trend detection, making advanced analytical capabilities available without manual model configuration.

Furthermore, hybrid cleaning systems that combine deterministic, rule-based logic with probabilistic machine learning models have been shown to outperform single-method approaches in both accuracy and user trust [27]. To preserve downstream data integrity, contemporary literature strongly advocates the automated optimization of multiple algorithmic strategies—including distance-based K-Nearest Neighbors (KNN) and iterative regression modeling—to seamlessly replace missing data values without introducing mathematical skew or cognitive burden on the user [28].

4. Automated Data Structuring and Summarization

Across the reviewed literature, a consistent gap emerges: while existing tools demonstrate technical efficacy, they largely target developers or data

scientists familiar with programming environments. Research indicates that the 'Quality at Source' principle is best achieved when individuals who understand the data's context—such as local researchers—are empowered to clean it [29]. Recent advancements have introduced “exception explainers” that provide AI-powered fix suggestions to bridge this gap [30], yet comprehensive, web-based systems that offer guided, step-by-step assistance specifically for non-technical users in academic settings remain underrepresented. Smartalyze addresses this gap by combining rule-based preprocessing and intuitive visualizations in a web-based interface designed specifically for students and researchers with limited technical backgrounds. To ensure maximum effectiveness, data visualization for non-experts must adhere to 'Minimalist Design' principles to prevent cognitive overload during the insight generation phase [31].

In summary, the literature underscores that while the technical backend for automated cleaning is robust, there is a significant lack of accessible interfaces for non-experts. Smartalyze fills this void by prioritizing usability alongside technical automation.

Related Works

The following systems represent existing approaches to automated data cleaning and analysis. While these systems offer powerful functionalities, Smartalyze differentiates itself by focusing on web-based accessibility for the DNSC community.

Feature / System	Smartalyze (Proposed)	Tale of Data	Open Refine	ML Data Cleaning Review (Yasin & Khorsheed)

Web-based interface	✓	✓		
Upload structured data (CSV/Excel/JSON)	✓	✓	✓	
OCR document text extraction	✓			
Automated data cleaning guidance	✓	✓	✓	✓
Machine learning used	✓			✓

Discussion of Related Works

Tale of Data (Data Cleansing Automation Tool)

Tale of Data operates as a no-code web platform designed to automate quality checks, validate complex business rules, and resolve data non-conformities without manual programming [32]. Similar to Smartalyze, it provides an accessible client interface for users to upload structured datasets and execute rule-based cleaning actions. However, Tale of Data focuses heavily

on deterministic corporate business rules and data compliance rather than deep machine learning trend prediction. Furthermore, it completely lacks an integrated Optical Character Recognition (OCR) text extraction layer, making it impossible for users to upload image-based documents for tabular dataset conversion.

OpenRefine

OpenRefine is a powerful, free, open-source tool dedicated to working with messy data through extensive cleaning, format transformation, and dataset extensions [33]. It features rich utilities for handling structured file types such as CSV, Excel, and JSON, offering algorithmic clustering to find text inconsistencies and hidden data errors. However, OpenRefine operates strictly as a local desktop software application rather than a cloud-hosted web platform, requiring manual installation and device-dependent storage. Furthermore, it completely lacks a native conversational AI assistant, omits front-end OCR document text extraction pipelines, and relies on its own complex domain expression language for sophisticated transformations, creating a steep entry barrier for data-illiterate users.

Automated Data Cleaning in Large Databases Using Machine Learning Methods (Yasin & Khorsheed)

The research framework designed by Yasin and Khorsheed explores automated data cleaning techniques inside large-scale database environments by integrating hybrid supervised and unsupervised machine learning algorithms [34]. Their methodology targets data volume and complexity issues, utilizing algorithmic workflows to improve overall data quality, accelerate data preparation efficiency, and reduce error-prone manual cleansing workloads. While their work demonstrates powerful backend analytical capabilities, it functions strictly as a deep, non-deployed technical research framework rather than public, self-service software. It lacks a cloud web interface, does not support direct user-facing uploads for local flat files (like custom Excel or JSON sheets), and completely omits an OCR layer for scanning static table text from static graphics.

Synthesis and Justification for Smartalyze

The comparative matrix demonstrates that while existing industry solutions and academic frameworks contain robust data processing, streaming, analytics, and validation capabilities, they remain deeply fragmented. Systems are either restricted to corporate compliance tracking, bound to expensive business data replication networks, designed as automated developer testing tools, built as heavy streaming data infrastructures, or deployed as standalone desktop clients and complex, non-visual engineering architectures.

Smartalyze bridges this gap by condensing web accessibility, automated rule-based cleaning, and machine learning trend analysis into a singular, zero-cost dashboard. Crucially, by adding an Optical Character Recognition (OCR) document text extraction layer, Smartalyze introduces a novel data ingestion avenue missing from all six alternatives. This enables non-technical students, faculty, and local researchers within the Davao del Norte State College community to easily digitize static table images, mitigate cognitive strain, and confidently accelerate their academic data operations.

Definition of Terms

Anomaly Detection. In this study, anomaly detection refers to the process by which the Smartalyze system automatically identifies data points or patterns that deviate significantly from the expected distribution within an uploaded dataset, using basic machine learning models to flag unusual values for user review.

Data Cleaning. As used in this project, data cleaning refers to the automated and guided process of detecting and correcting errors, inconsistencies, duplicate

records, and missing values in a structured dataset to improve its accuracy and usability for analysis.

Data Preprocessing. In the context of this system, data preprocessing encompasses all transformations applied to a raw dataset prior to analysis, including data cleaning, type conversion, normalization, sorting, grouping, and summarization, performed to make the data suitable for machine learning or statistical analysis.

Machine Learning. For the purposes of this study, machine learning refers to the computational techniques employed by Smartalyze to automatically detect trends and anomalies in user-uploaded datasets, without requiring the user to manually configure or train models.

Rule-Based Preprocessing. As applied in this system, rule-based preprocessing refers to a set of predefined logical conditions and operations that Smartalyze uses to automatically detect and address common data quality issues, such as missing values, duplicate rows, and inconsistent data types, based on established data cleaning heuristics rather than learned patterns.

Smartalyze. Smartalyze is the AI-assisted, web-based data-cleaning and analysis system developed for this capstone project. The name is a portmanteau of “smart” and “analyze,” reflecting the system’s goal of enabling intelligent, guided analysis of structured datasets for non-technical users.

Structured Dataset. In this study, a structured dataset is data organized in a clearly defined tabular format, with rows representing individual records and columns representing distinct attributes or variables, such as files in CSV, Excel, or JSON formats that conform to a consistent schema.

Trend Detection. As used in this project, trend detection refers to the system’s ability to identify patterns of consistent increases, decreases, or cyclical behavior across data columns over time or across categories, enabling users to draw meaningful conclusions about the direction or tendencies within their dataset.

Visualization. In this system, visualization refers to the graphical representation of processed data through charts, tables, and diagrams generated by Smartalyze to help users intuitively understand patterns, trends, and anomalies identified during the analysis phase.

Web-Based System. For this study, a web-based system is an application accessed through a web browser, without requiring local software installation, enabling users to interact with its features on any device with internet connectivity.

Chapter 2

CHAPTER 2

METHODOLOGY

The development of Smartalyze follows the Iterative Software Development Model, a practical and widely used approach to system development in which software is built and improved through repeated cycles. As illustrated in Figure 1, each iteration involves structured, continuous phases: planning, analysis, design, implementation, testing, and evaluation, allowing for the strategic refinement of features based on incremental testing metrics and direct user feedback.

This methodology is highly suitable for Smartalyze because the system involves evolving components, such as data preprocessing rules, visualization outputs, and machine-learning-based anomaly detection. Since these features require continuous improvement and validation using different datasets, an iterative approach ensures flexibility and gradual enhancement of system performance and usability.

Each iteration of development produces a functional version of the system, starting with basic dataset upload and cleaning and gradually expanding to include advanced features such as automated data analysis, anomaly detection, and exportable insights, as outlined in the cyclic framework of Figure 1.

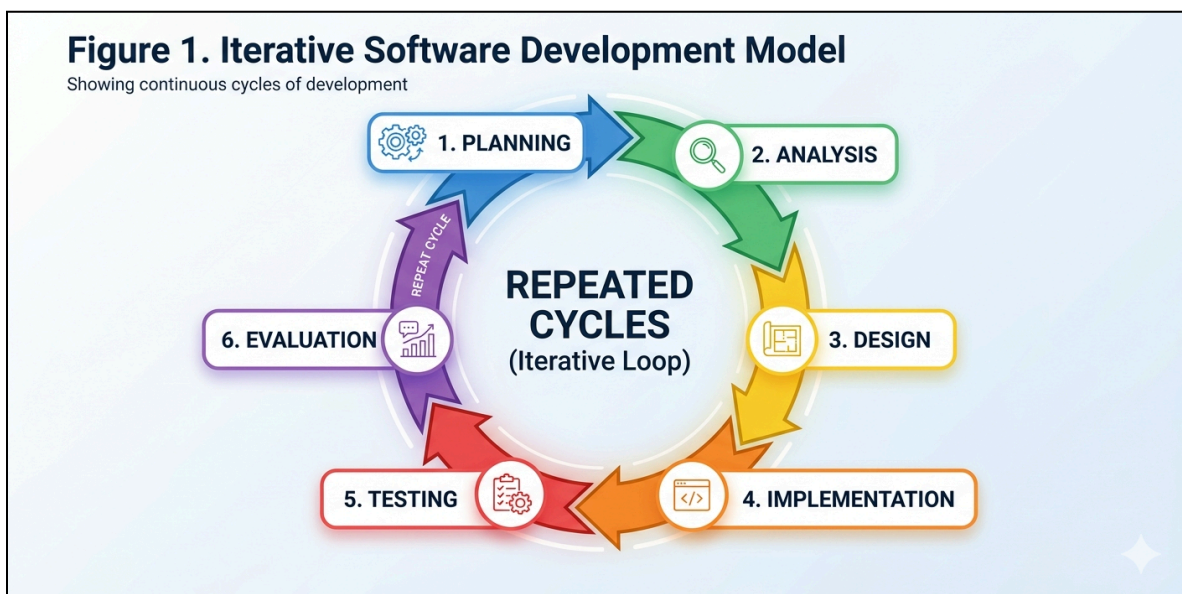


Figure 1. Iterative Software Development Lifecycle

Iteration Planning

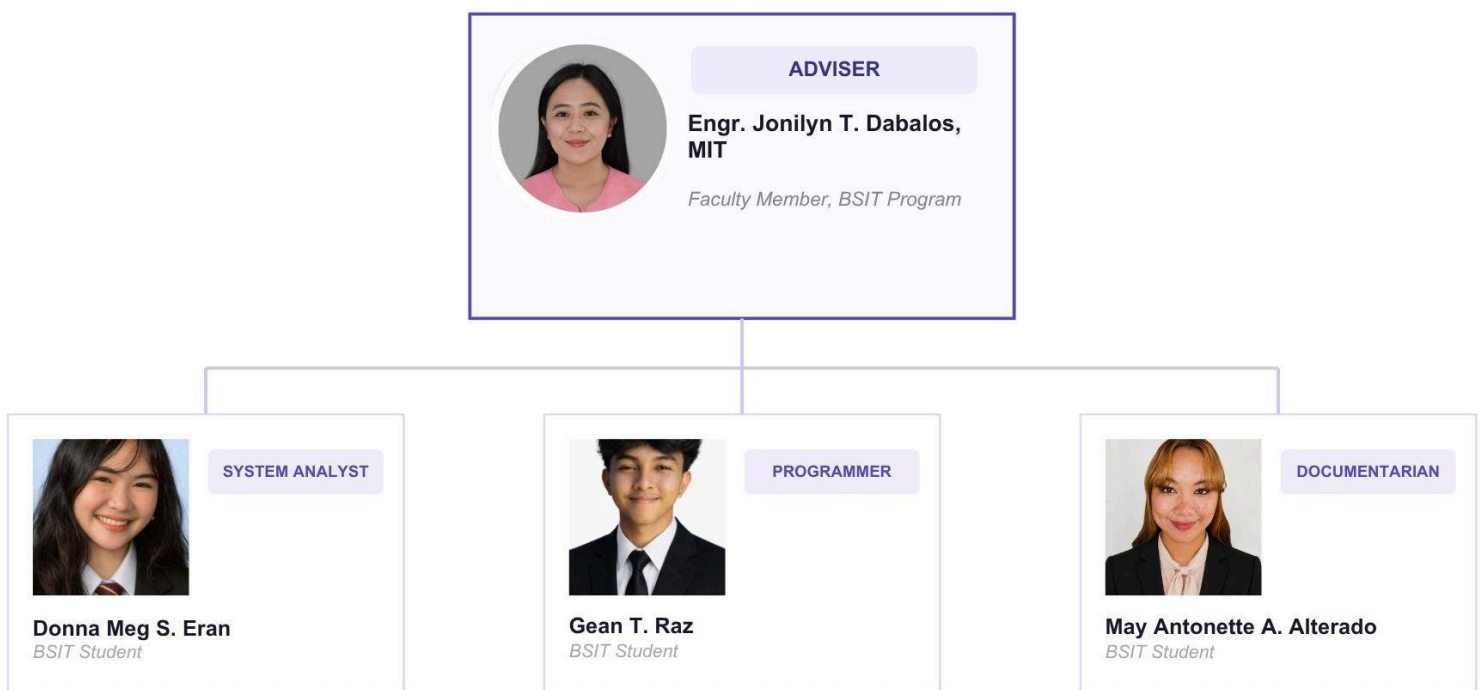
The conceptualization of Smartalyze stemmed from the observation of a critical gap in data literacy and management within the local academic and business communities at Davao del Norte State College. While data-driven decision-making is increasingly necessary, many students, researchers, and small business owners lack the programming expertise required to process structured datasets efficiently [35]. Traditional manual data preprocessing is repetitive, time-consuming, and error-prone, often consuming a significant portion of the analytical workflow [36]. Recognizing this challenge, the project was proposed to bridge the technical divide by automating routine data cleaning and basic data analysis tasks.

Under the Iterative Software Development Model, the planning process is conducted at the beginning of each development cycle. Instead of defining all requirements at once, the system is developed incrementally, allowing features to evolve based on testing results and user feedback [37]. In the initial iteration, the primary objective was established: to develop an AI-assisted web-based system that integrates rule-based preprocessing with basic machine learning techniques for trend analysis and anomaly detection [38].

The system's scope was defined to handle structured datasets in formats such as CSV, Excel, and JSON. Additionally, feasibility considerations were evaluated, including the use of Python-based data processing pipelines, web-based system architecture, and resource allocation for development [39]. Recent evaluations of modular web architectures confirm that Python-based frameworks are highly effective at democratizing data science by enabling the seamless integration of automated cleaning libraries into user-friendly interfaces [40]. A flexible development timeline was also established, allowing iterative refinement of system components, including data cleaning, analysis, and visualization modules, throughout successive development cycles.

Project Team Organization

The organizational structure is designed to allocate specific responsibilities essential for the successful development and deployment of Smartalyze. The team follows a collaborative hierarchy in which technical execution is balanced with rigorous



Project Adviser

The Project Adviser provides expert guidance and supervision throughout the development of the Capstone Project. She ensures that the project aligns with academic standards and technical requirements. The adviser reviews the system design, validates methodologies, and provides recommendations to enhance the system's overall quality and reliability.

Systems Analyst

The Systems Analyst, Donna Meg S. Eran, is responsible for analyzing system requirements and ensuring that all system components align with the project objectives. She evaluates user needs, defines system functionalities, and ensures consistency across all modules. The Systems Analyst also coordinates with the development team to ensure that the system design is properly implemented and meets the specific demands of non-technical stakeholders.

System Programmer

The System Programmer, Gean T. Raz, is responsible for the actual development and implementation of the system. He translates system designs into functional code, develops system features, integrates modules, and performs debugging and testing. He ensures that the system operates efficiently, handles various data formats accurately, and meets the required technical specifications established during the planning phase.

Documentarian / Technical Writer

The Documentarian, May Antonette Alterado, is responsible for preparing and organizing all project documentation. This includes the system documentation, research manuscript, and technical reports. She ensures that all documents are accurate, well-structured, and aligned with academic and technical standards. This role is vital for maintaining a clear record of the development lifecycle and providing a comprehensive guide for future system maintenance.

Work Breakdown Structure (WBS)

The project activities for Smartalyze are organized according to the Iterative Software Development Model, in which the system is developed through incremental cycles. Each iteration consists of planning, analysis, design, implementation, and testing, allowing continuous refinement of system features.

Iteration 1 – Core System Setup: Focuses on foundational project planning, requirements gathering for basic dataset formats (CSV, Excel, JSON), designing the client-server architecture, and implementing the core web features for file uploading and database viewing.

Iteration 2 – Data Cleaning Module: Centers on analyzing common data anomalies (such as missing values, duplicates, and faulty data types) and building the rule-based backend preprocessing engine to detect these issues and offer automated fix suggestions to the user.

Iteration 3 – Analysis and Intelligence Layer: Integrates machine learning capabilities into the application to automatically identify deeper data patterns, detect anomalies, and generate trend insights from the uploaded datasets.

Iteration 4 – Visualization and Reporting: Focuses on the user experience by building minimalist chart layouts, graphical elements, and data summaries, as well as implementing reporting features that allow users to export their refined data.

Iteration 5 – System Testing and Refinement: Dedicates final efforts to overall platform optimization, bug fixing, performance bottleneck resolution, and comprehensive quality assurance via unit, integration, and user acceptance testing (UAT).

Gantt Chart

The project schedule and operational timeline for the tasks defined in the Work Breakdown Structure (WBS) are mapped out chronologically in Figure 3. This timeline illustrates the sequential progression of the iterative development phases across their respective calendar weeks, leading to the targeted completion in May 2026. As shown in the visual breakdown of Figure 3, each system module transitions dynamically into the next to ensure milestone accountability throughout the project lifecycle.

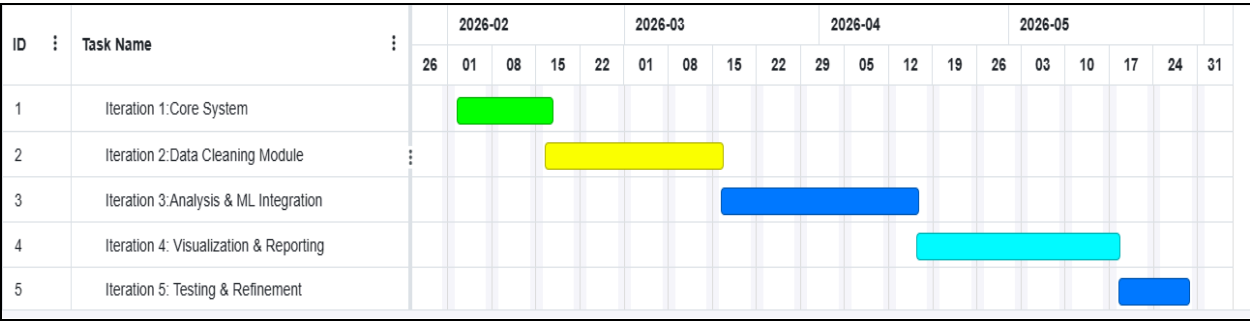


Figure 3: Gantt Chart

System Analysis

This section presents the analysis of the Smartalyze system, focusing on how user requirements are translated into system functionality. It defines the conceptual framework, identifies system requirements, and illustrates the logical flow of data as users interact with the platform. The goal of this phase is to ensure that raw structured datasets can be efficiently transformed into meaningful insights for non-technical users.

In alignment with the Iterative Software Development Model, system analysis is continuously refined across development cycles to improve data processing, analysis accuracy, and user interaction.

System Architecture

Smartalyze follows a web-based client-server architecture, enabling users to access the system through a standard web browser without requiring local installation. This architecture supports scalability, accessibility, and centralized data processing.

The system is composed of the following components:

Hardware Components

The system operates on a central server that handles data processing and computation. End users access the platform through client devices such as desktop computers or laptops with internet connectivity.

Software Components

The frontend is implemented as a browser-based user interface designed for intuitive interaction, allowing users to upload datasets, view results, and perform operations without coding knowledge.

The backend uses Python-based data processing libraries to perform rule-based data cleaning and apply basic machine learning algorithms for trend analysis and anomaly detection.

Data Components

The system processes structured datasets provided by users in formats such as CSV, Excel, and JSON. These datasets undergo validation, preprocessing, analysis, and transformation within the system pipeline.

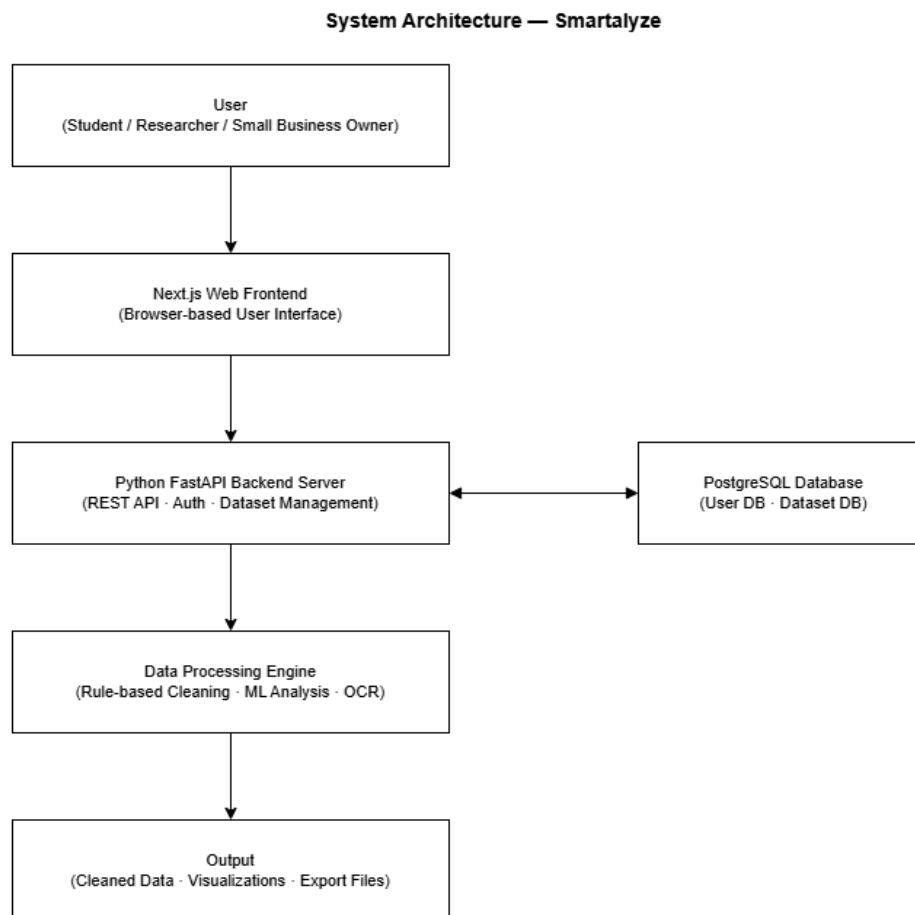


Figure 4: Smartalyze System Architecture

Functional and Non-Functional Requirements

Functional Requirements

These define the core capabilities of the system:

1. The system must allow users to upload structured datasets in CSV, Excel, or JSON formats.
2. The system must automatically detect data quality issues such as missing values, duplicate records, and inconsistent data types.
3. The system must provide guided suggestions for dataset structuring, including grouping, sorting, and summarization.
4. The system must apply basic machine learning techniques to identify trends and detect anomalies.
5. The system must generate visualizations such as charts and tables to present insights clearly.
6. The system must allow users to export cleaned datasets and analysis results for external use.

Non-Functional Requirements

These define the quality attributes of the system:

1. **Usability:** The system must provide a simple and intuitive interface tailored for non-technical users, minimizing the need for prior programming knowledge.
2. **Performance:** The system should process small to medium-sized datasets efficiently. Processing delays for large datasets are acceptable within the limitations of a web-based prototype.
3. **Reliability:** The system must produce consistent and accurate results when processing the same dataset under the same conditions.
4. **Privacy and Scope:** The system must only process datasets explicitly uploaded by the user and must not perform any background data collection or real-time monitoring.

5. **Accessibility:** The system must be accessible through modern web browsers across different operating systems.
6. **Maintainability:** The system should be modular, enabling updates to preprocessing rules and machine learning components without affecting the rest of the system.

Use Case Diagram

The functional interactions between the user and the platform components are modeled in **Figure 5**, which outlines how the system boundaries respond to different behavioral actions.

- **Primary Actor:** Non-technical User (Student, Researcher, Small Business Owner).
- **Key Use Cases:** As mapped chronologically within the system boundary in **Figure 5**, the primary actor initiates core behaviors, including Upload Dataset, View Dataset, Apply Cleaning Operations, Run Data Analysis, and Generate Report. These functional entry points allow the user to manage behavioral iterations seamlessly without manual coding.

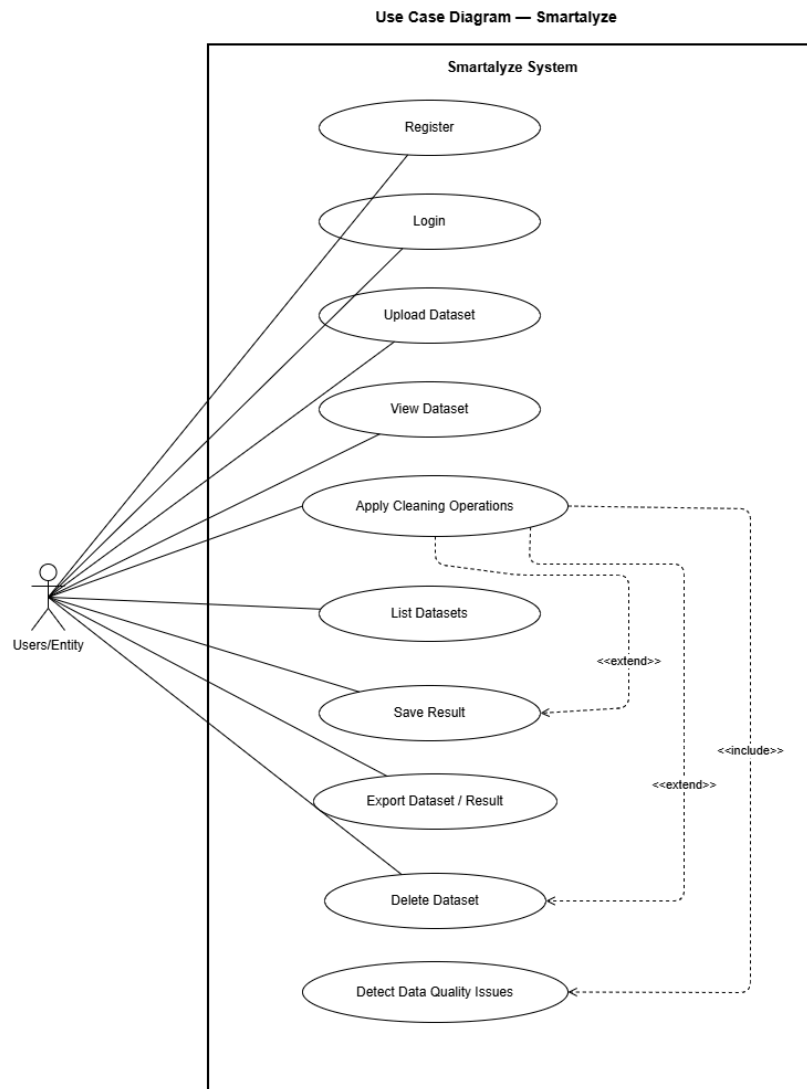


Figure 5: Use Case Diagram

Context Flow Diagram

Following the Gane and Sarson method, the Level 0 DFD mapped in Figure 6 illustrates the logical flow of business actions between the user and the entire Smartalyze system as a single process.

External Entity: User

Process 0: Smartalyze System

Data Inflow: Raw Structured Dataset File (CSV/Excel/JSON), Requests

Data Outflow: Cleaning Results, Exported Dataset

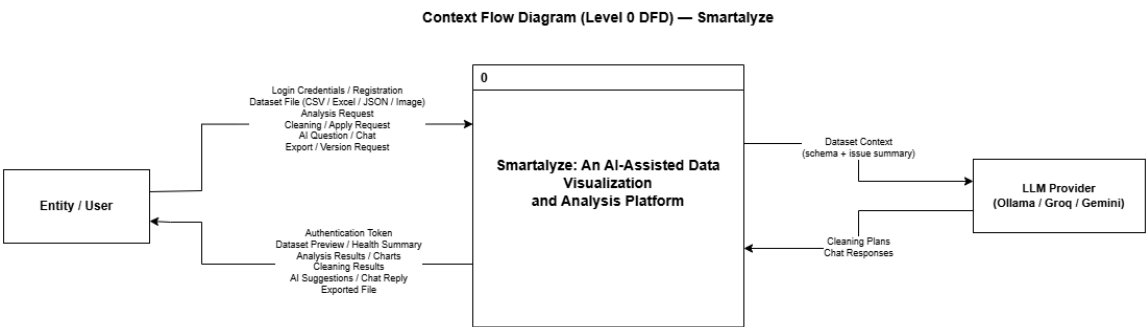


Figure 6: Level 0 (Context Flow Diagram)

Level 1 Data Flow Diagram

The Level 1 Data Flow Diagram (DFD) presents a detailed decomposition of the Smartalyze system, expanding the single process shown in the Level 0 diagram into multiple interconnected sub-processes. This representation follows the Gane and Sarson method and illustrates how data moves through the system—from user input to system outputs—while interacting with the system’s data stores.

The diagram highlights the interaction between the user, the Smartalyze sub-processes, and the primary data stores: the User Database (accounts and sessions) and the Dataset Database (datasets, snapshot versions, and action logs).

Smartalyze stores dataset contents as versioned JSON snapshots and returns cleaning results and export files based on the current snapshot or a user-provided one.

Processes:

1.0 User Authentication: Manages user registration, login verification, and secure session management by validating credentials and issuing authentication tokens.

2.0 Dataset Workspace / Dataset Management: Handles the uploading, structural validation, previewing, listing, and deletion of structured dataset files (CSV, Excel, JSON).

3.0 Data Cleaning (Detect and Apply): Executes rule-based anomaly detection (missing values, duplicates, bad data types) on a dataset snapshot and applies on-the-fly fixes without immediately overwriting the permanent database.

4.0 Export Dataset/Result: Generates and serves a downloadable file (CSV, XLSX, or JSON) from either a permanently stored dataset snapshot or an unsaved, active cleaning workspace result.

5.0 Save Result: Commits cleaned data snapshots to permanent storage, allowing the user to either version-control and overwrite the existing file or save the output as an entirely new dataset.

Data Stores (Referenced in the Level 1 DFD):

D1 User Database: Accommodates account profiles, credential hashes, active session states, and security token validations.

D2 Dataset Database: Maintains the core data architecture by storing dataset metadata, file contents across versioned snapshots, and comprehensive audit trails/action logs.

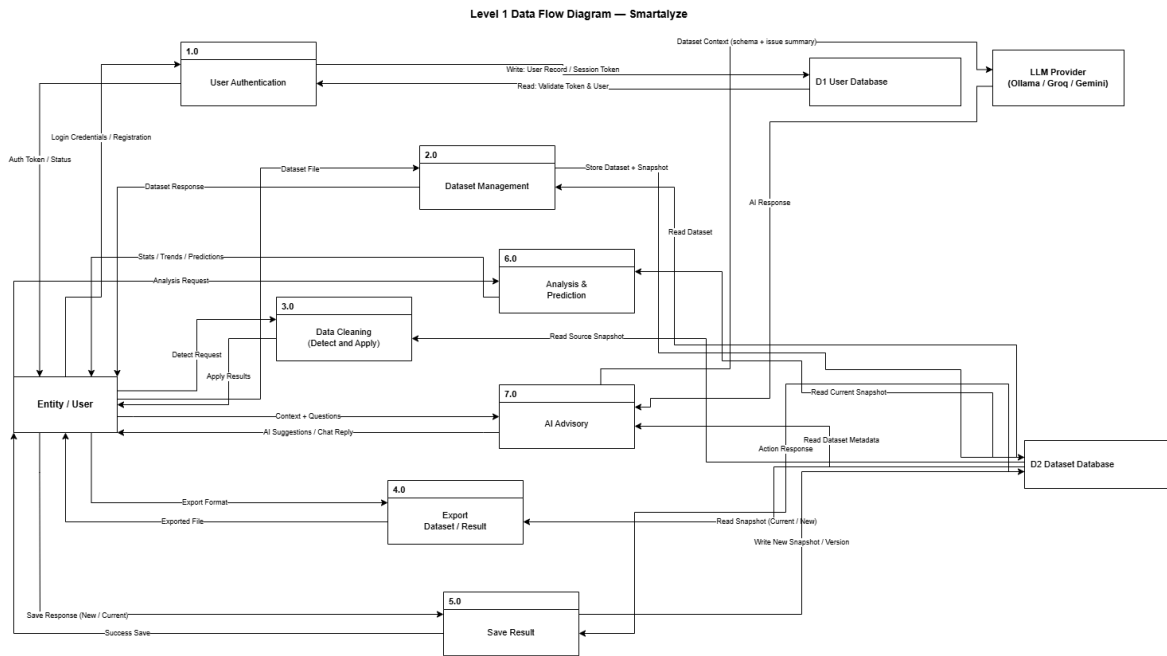


Figure 7: Level 1 (Data Flow Diagram)

System Design

This section presents the structural design artifacts that define Smartalyze's database schema, relationships, and technical blueprint. These components ensure that the data model supports all functional requirements identified during the analysis phase.

Entity Relationship Diagram (ERD)

The Entity-Relationship Diagram depicts the logical structure of the database, showing the entities in the Smartalyze system and the relationships between them. It maps out how datasets, users, preprocessing sessions, and analysis results are interconnected.

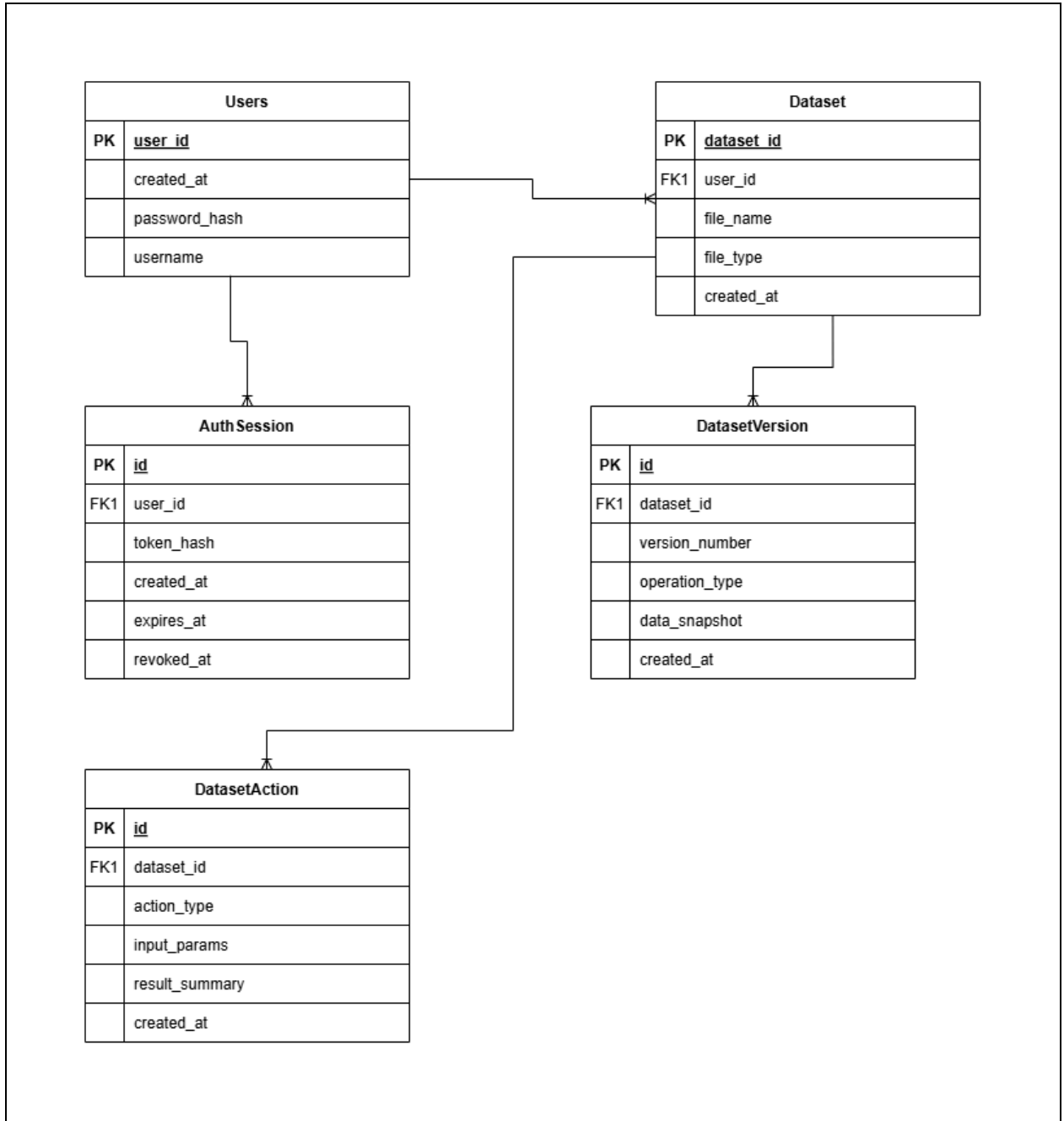


Figure 8: ERD (Entity-Relationship Diagram)

Data Dictionary

The Data Dictionary provides a structured reference of all data elements used within the Smartalyze system. It defines the attributes, data types, key types, and descriptions of each data entity stored in the database. This ensures consistency in data handling and supports efficient system development and maintenance.

Table 1.1 Users

Attribute	Data Type	Key Type	Description
user_id	Integer	Primary Key	Unique identifier for each user
username	Varchar	—	Username used for login
password_hash	Varchar	Unique	Hashed password for auth
created_at	Timestamp	—	Date and time the user account was created

Table 1.2 AuthSessions

Attribute	Data Type	Key Type	Description
user_id	Integer	Foreign Key	References the user who owns the session
token_hash	Varchar	unique	Hashed authentication token (Bearer token stored securely)
created_at	Timestamp	—	Data and time the session is created
expires_at	Timestamp	—	Date and time the session expires
revoked_at	Timestamp	—	Date and time the session was revoked (null if active)

Table 1.3 Datasets

Attribute	Data Type	Key Type	Description
id	Integer	Primary Key	Unique identifier for each dataset
user_id	Integer	Foreign Key	References the user who owns the dataset
name	varchar	—	Dataset name

file_type	varchar	—	Dataset format
description	text	—	Optional user dataset desc
current_version_id	integer	Foreign Key	References to the current dataset snapshot
created_at	Timestamp	—	Date and time the dataset was created/uploaded

Table 1.4 Dataset Versions

Attribute	Data Type	Key Type	Description
id	Integer	Primary Key	Unique identifier for each dataset snapshot/version
dataset_id	Integer	Foreign Key	References the dataset this snapshot belongs to
version_number	integer	—	Version sequence number within a dataset like 1 2 3
operation_type	varchar	—	Label for how the snapshot was created (e.g., upload, cleaning, result)
data_snapshot	JSONB	—	Full stored snapshot of the dataset content, including records/preview/summary

created_at	Timestamp	—	Date and time the snapshot was created
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Table 1.5 Dataset_Actions

Attribute	Data Type	Key Type	Description
id	Integer	Primary Key	Unique identifier for each dataset action event
dataset_id	Integer	Foreign Key	References the dataset affected by the action (datasets.id)
action_type	varchar	—	Type of action performed (e.g., upload, result, delete)
input_params	JSONB	—	Input parameters used for the action (stored as JSON)
result_summary	JSONB	—	Summary of outputs/results for the action (stored as JSON)
created_at	Timestamp	—	Date and time the action was recorded

Technologies, Concepts, and Theories

This section presents the core technologies, concepts, and data processing techniques implemented in Smartalyze. The system integrates rule-based preprocessing and basic machine learning methods to transform raw structured datasets into meaningful insights. The following subsections describe the key processes involved in the system pipeline.

Data Collection

Data collection in Smartalyze is user-driven. The system does not gather data from external sources; instead, users upload structured datasets in formats such as CSV, Excel, and JSON.

Upon upload, datasets are validated to ensure correct formatting and compatibility with the system. This includes checking file structure, column consistency, and data readability. The collected data is then stored temporarily in the system for further processing.

This approach ensures that the system remains domain-independent, allowing users from different fields (academic, business, research) to analyze their own datasets.

Data Pre-processing

Data preprocessing is a critical step in the Smartalyze system, as raw datasets often contain inconsistencies that affect analysis accuracy.

The system applies rule-based preprocessing techniques, including:

- **Handling Missing Values:** Detecting null or empty fields and applying appropriate strategies such as removal or imputation
- **Duplicate Detection:** Identifying and removing repeated records

- **Data Type Validation:** Ensuring values match expected formats (e.g., numeric, categorical, date)
- **Data Cleaning Suggestions:** Providing guided recommendations for improving dataset quality

This automated preprocessing reduces manual effort and prepares the dataset for reliable analysis.

Feature Extraction

Feature extraction involves identifying and transforming relevant data attributes to make them suitable for analysis.

In Smartalyze, this process includes:

- **Column Type Identification:** Classifying attributes as numerical, categorical, or datetime
- **Aggregation and Grouping:** Summarizing data using operations such as count, sum, and average
- **Derived Features:** Generating new variables (e.g., totals, differences, trends over time)
- **Data Transformation:** Normalizing or restructuring data for analysis

These steps enhance the quality and interpretability of the dataset, allowing the system to generate more meaningful insights.

Classification and Analysis

Smartalyze applies basic analytical and machine learning techniques to interpret the processed dataset.

Rather than complex predictive modeling, the system focuses on practical and interpretable methods, including:

- **Trend Analysis:** Identifying patterns over time or across categories

- **Anomaly Detection:** Detecting outliers or unusual data points using statistical methods (e.g., z-score, threshold-based detection)
- **Data Classification (Basic):** Grouping data into categories based on attribute values

These techniques enable users to understand their data without requiring advanced knowledge in machine learning.

Rule-Based Preprocessing

Rule-based preprocessing refers to applying predefined logical conditions to identify and resolve data quality issues. For Smartalyze, this includes automated detection and handling of missing values, duplicate entries, and data type inconsistencies within uploaded structured datasets.

Machine Learning for Anomaly Detection

Basic machine learning models are employed to identify patterns within cleaned datasets. These models flag data points that deviate significantly from established trends, enabling automated anomaly detection without manual inspection.

System Testing and Implementation

This section describes the testing strategies and implementation approach used to validate the functionality, performance, and usability of the Smartalyze system. A multi-layered testing methodology was employed to ensure that each component meets the requirements defined in the analysis phase.

Unit testing was conducted on individual backend modules to verify that each data-processing function—such as missing-value detection, duplicate removal, and type normalization—performs correctly in isolation [41]. Integration testing follows to ensure

that all system components, including the frontend interface, the backend processing engine, and the machine learning modules, interact seamlessly within the unified architecture [42]. Finally, pilot testing was conducted with a sample of non-technical users to evaluate the system's usability and gather feedback for refinement, ensuring the tool effectively addresses the data literacy challenges of the target audience [43].

System Maintenance

System maintenance encompasses the ongoing activities required to keep Smartalyze operational, reliable, and aligned with user needs following its initial deployment. This phase ensures that the system remains functional and continues to meet its stated objectives over time.

Performance monitoring is conducted regularly to identify bottlenecks, particularly for users processing very large datasets. Based on collected performance metrics, optimization strategies are applied to reduce processing time and improve system responsiveness. Additionally, rule-based heuristics and machine learning models are periodically updated with user feedback to enhance the accuracy of data quality detection and anomaly identification, ensuring the system evolves to meet its users' requirements.

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Tab 3

The increasing volume and complexity of data in modern enterprises have spurred research into automated data preprocessing and analysis tools [16]. A growing body of literature addresses the challenge of making data science accessible to non-technical users by eliminating the need for manual coding or specialized expertise [17]. This section reviews the key themes, gaps, and relevant works that inform the design of Smartalyze.

1. Data Analytics and Data-Driven Decision Making

The contemporary academic and institutional landscape increasingly relies on data-driven decision-making to optimize performance and identify operational insights. This paradigm shift marks a clear transition away from a reliance on subjective intuition or anecdotal evidence toward organizational choices that are strictly grounded in objective dataset metrics. However, extracting real value from these datasets often requires advanced analytical competencies that non-technical users—such as local students, educators, and business researchers—frequently do not possess.

Literature emphasizes that when analytical frameworks require specialized scripting or complex database environments, non-technical users experience structural barriers, causing notable frustration and frequently leading them to abandon data-driven approaches entirely [27]. To bridge this gap, modern research heavily advocates for "data democratization," which is defined as the intentional broadening of data access and analytical tools to non-specialist stakeholders within an organization [28]. This movement emphasizes the deployment of self-service analytics interfaces that minimize cognitive load, effectively allowing individuals without programming backgrounds to comfortably cleanse, visualize, and interpret data trends to make informed decisions [29].

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2. Data Cleaning and Data Preprocessing

Research on automated data preprocessing consistently highlights the bottleneck posed by manual data cleaning. Statistical evidence indicates that data preparation accounts for nearly 80% of the analytical lifecycle, underscoring that operations such as handling missing values

and normalizing inconsistent formats are foundational yet time-intensive and prone to human error [18].

Furthermore, contemporary evaluations demonstrate that Python-based automation pipelines can significantly accelerate these workflows, reducing manual effort and ensuring reproducible results [19]. Recent benchmark studies on large real-world datasets confirm that these automated tools can scale effectively without sacrificing data quality [20]. To optimize these pipelines, recent studies emphasize that integrating automated anomaly detection algorithms helps identify structural data irregularities early, dramatically reducing the manual scripting required by end-users [21]. These works collectively establish the feasibility and value of automating data cleaning, forming the theoretical basis for Smartalyze's rule-based engine.

[18] (Your original citation for the 80% lifecycle data)

[19] (Your original citation for Python-based automation)

[20] (Your original citation for benchmark studies)

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3. Missing Data Handling Techniques

The integration of machine learning (ML) into data cleaning has received growing scholarly attention. New frameworks demonstrate that ML methods can effectively identify and resolve quality issues, particularly in anomaly detection and missing-value imputation [22]. Recent tools, such as *cleanTS*, extend this approach to time-series data, demonstrating that systems that leverage statistical models can perform domain-specific cleaning with minimal user input [23]. Smartalyze draws from these studies by incorporating basic ML models for trend detection, making advanced analytical capabilities available without manual model configuration.

Furthermore, hybrid cleaning systems that combine deterministic, rule-based logic with probabilistic machine learning models have been shown to outperform single-method approaches in both accuracy and user trust [24]. To preserve downstream data integrity, contemporary literature strongly advocates the automated optimization of multiple algorithmic strategies—including distance-based K-Nearest Neighbors (KNN) and iterative regression modeling—to seamlessly replace missing data values without introducing mathematical skew or cognitive burden on the user [25].

[22] (Your original citation for ML methods in data cleaning)

[23] (Your original citation for *cleanTS*)

[24] (Your original citation for hybrid cleaning systems)

[25] M. Irfan, S. Patabandige, and R. Sajanraj, "Machine learning-based COD prediction enhancement in microalgae cultivation with POME using imputation techniques," *IEEE Transactions on Big Data Engineering*, vol. 12, no. 2, pp. 451–458, Mar. 2026. Available: <https://lbeifits.wordpress.com/wp-content/uploads/2026/03/irfan-m.pdf>

4. Automated Data Structuring and Summarization

Across the reviewed literature, a consistent gap emerges: while existing tools demonstrate technical efficacy, they largely target developers or data scientists familiar with programming environments. Research indicates that the 'Quality at Source' principle is best achieved when the individuals who understand the data's context—such as local researchers—are the ones empowered to clean it. Recent advancements have introduced "exception explainers" that provide AI-powered fix suggestions to bridge this gap, yet comprehensive, web-based systems that offer guided, step-by-step assistance specifically for non-technical users in academic settings remain underrepresented. Smartalyze addresses this gap by combining rule-based preprocessing and intuitive visualizations in a web-based interface designed specifically for students and researchers with limited technical backgrounds. To ensure maximum effectiveness, data visualization for non-experts must adhere to 'Minimalist Design' principles to prevent cognitive overload during the insight generation phase.

Furthermore, optimizing the user experience requires the system to handle the immediate extraction of data patterns. Recent studies highlight that automated data profiling architectures can successfully scan raw datasets, infer underlying schemas, and generate high-level structural summaries without manual intervention [26]. Providing these automated data definitions and distribution summaries directly to non-technical users significantly decreases the technical barrier to entry, allowing them to instantly comprehend dataset frameworks before diving into deeper analytics [26]. In summary, the literature underscores that while the technical backend for automated cleaning is robust, there is a significant lack of accessible interfaces for non-experts. Smartalyze fills this void by prioritizing usability alongside technical automation.